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Chapter 4

Inventory Updates



Inventory Updates Introduction

The investment in Inventory typically represents one of the largest single areas of investment in a business, often as high as 25% of total assets. It is vital therefore that planning and control of its acquisition and disposition are exercised.

A number of techniques have been developed to assist in this control. In the sphere of a Production operation where demand is considered to be dependent, the most common forms of control are Material Requirements Planning (MRP), or Just In Time (JIT).

These techniques work in the relatively predictable environment where demand is dependent on an agreed production schedule. Such control techniques are provided in the Production Applications.

However, the demand for end-product items is influenced primarily by factors that are independent of company decisions. These external factors include a certain amount of random variation in the demand for such items. As a result, forecasts of demand for these items are typically projections of historical demand patterns to determine average usage.

The concepts of independent and dependent demand are important in selecting appropriate Inventory Management techniques.

There are generally considered to be two types of Inventory: Movement and Organisational.

Inventory Holding

- Work In Progress - in a Production organisation.
- Stock In Transit - in a Distribution environment.
- Cycle Stock is a result of buying or making more than is required for the immediate requirement.
- To determine the Economic Order Quantity, you must set the one-time costs of raising an order and perhaps setting machines or transport against the costs of holding stock.
- You would hold anticipation stock to deal with times when demand exceeds the rate at which the item may be supplied economically.
- Highly seasonal products such as lawn mowers are a good example of where it is appropriate to have a fairly steady rate of production throughout the year, and build high levels of stock over the winter so that the depletion rate of stock plus the production rate equals demand during high summer.

Inventory Decisions

You need to make two types of routine decisions when managing inventories. These decisions are concerned with how much and when to order. Four main factors influence these decisions:

- Forecast demand.
- Replenishment lead time.
- Inventory related cost.
- Management policies.

There are two important parameters to assist inventory controllers in making these decisions, they are:

- Economic Order Quantity - EOQ.
- Economic Time Between Orders - ETBO.

The standard calculation of EOQs:

$$EOQ = \sqrt{(2 * CP * AU) / CH}$$

Where:

CP = Cost of order placement.

AU = Average usage per annum.

CH = Inventory holding cost per year.

$$CH = \text{Item Cost} * \text{Carrying Rate} / 100$$

Once EOQ has been determined, then:

$$ETBO = EOQ / WU.$$

Where WU = Weekly Usage.

The resulting Economic Time Between Orders (ETBO) is in weeks.



Note: The cost of order placement and carrying rate are both specified on the Inventory company profile.

Demand Management

Inventory Management provides a range of facilities that contribute to the management of independent demand items, traditionally controlled by reorder point techniques.

- A value/usage matrix lets you implement a stock investment policy at each stockroom.
- Usage profiles provide a range of calculations for forecast usage.
- EOQ calculation is performed and can be used as the basis of reorder quantity and time between orders.

- Lead time can be factored in the reorder calculation to modify order review periods.

You can use the Inventory Management system to make re-order level calculations.

Many different criteria can be used within the calculations to reflect differing item usages, values and stock control requirements.

The culmination of the process is the production of the Reorder/Overstock report that details all recommendations.

Included in the prior set up are:

- The calculation of an Average Usage figure.
- The calculation of Reorder Point (ROP), Maximum Stock and Economic Order Quantity (EOQ).
- The selection of one of 5 Reorder Policies appropriate to each item.

Selection Of The Reorder Policy

The following data is used in the examples for each of the Reorder Recommendations:

Physical/Available	= 0
Expected	= -90
Max weeks = 8 Average Usage	= 10
Min weeks = 6 Lead Time	= 4 weeks
Reorder Factor	= 200%
ROP=(Min. Weeks) 6 + (Lt) 4 * (av. usage) 10	= 100
Maximum stock=(Max. Weeks) 8 * (av. usage) 10	= 80

Maximum stock is below ROP because the lead time (4) is greater than the difference between Maximum (8) and Minimum (6) weeks, in other words 2.

Reorder Policy 1 (Up To Max)

ROQ = Maximum stock - Expected + (average usage * lead time).

If you do not specify maximum stock on the item/stockroom record the appropriate value from the stock weeks grid is obtained. This is expressed in weeks and is multiplied by average usage to give a maximum stock value.

Example

$$\begin{array}{rcccccc} 80 & - & -90 & + & 40 & = & 210 \\ \text{Maximum stock} & - & \text{Expected} & + & (\text{av. usage} * \text{Lt}) & & \end{array}$$

Reorder Policy 2 (ROQ)

ROQ = Average Usage * lead time * FACTOR percentage.

Example

$$10 * 4 * 200\% = 80$$

$$\text{Av. usage} * (\text{lt}) * \text{Factor}$$

Reorder Policy 3 (Up To Min)

Calculate EOQ, etc. must be set to 0.

A ROP may be manually entered.

ROQ = ROP - Expected.

Example

$$0 - -90 = 90$$

Reorder Point - Expected

Calculate EOQ, etc. is set to 0 for this policy, therefore Inventory does not calculate ROP, it remains as 0 on the item/stockroom record.

Reorder Policy 4 (Up To ROP)

ROQ = ROP - Expected + (average usage * lead time).

Example

$$100 - -90 + 40 = 230$$

Reorder Point - Expected + (lt)

Reorder Policy 5 (EOQ)

ROQ = EOQ.

In Policies 1 to 4 you can manually enter a Reorder Quantity. Policy 3 is designed to reorder to demand. The stock position is reviewed as follows for all items:

1. Available stock is calculated as: Physical - Frozen - Allocated = Available.
2. Expected stock is calculated as: Available + Order + In Transit - Back Order - Reserved = Expected.
3. If expected stock is greater than maximum stock the item is reported as in Excess.
4. If expected stock is less than reorder point, the item is reported for Order action.
5. If order action is required, the reorder quantity is determined. If you manually specify a reorder quantity, it is recommended whatever the reorder policy. If not, the reorder quantity is determined for each policy as follows:
 - If the policy is Up to MAX, the system will recommend reorder up to the maximum stock holding.
 - If the policy is ROP, the system calculates a reorder quantity based on the reorder quantity factor, the average usage and the lead time or the difference between Expected and ROP if greater.
 - If the policy is Up to MIN, the system will recommend reorder up to the minimum stock holding.
 - If the policy is Up to ROP the system will recommend that you order sufficient to bring expected back to ROP.

- If the policy is EOQ, the system will recommend reorder of the calculated or maintained economic order quantity.

Example

If the Maximum Stock is:

*Maximum Stock Weeks * Average Usage.*

The Reorder Point is:

*(Minimum Stock Weeks + Lead Time (in weeks)) * Average Usage.*

The Expected Stock is:

Available + On Order + In Transit - Back Orders - Reserved.

The EOQ depends on Cost of Order placement, Average Usage per annum and Inventory Holding Cost per year.

Regular Updates

To improve the speed and quality of information displayed within the Inventory Management application a series of weekly and periodic balances are maintained. These balances need to be updated at week, month and year end.

At month end, both week end and month end should be run consecutively. The month end run normally includes a valuation of stock movements and this, therefore, is prompted.

At year end, week end, month end and year end should be run consecutively.

These batch options are accessed by selecting Updates. They are described in detail at each of the Update options.

Week End Update



1/INC

This batch option performs the following tasks:

- If the usage cycle set on the company profile is W, for weekly, then average weekly usage will be recalculated for each product/stockroom.
- Both the physical and in transit stock at the end of the week are stored in a separate field Stock At Start Of Week for each product/stockroom.
- Usage this week is set to zero for each product/stockroom.
- The current week number on the calendar file is incremented by 1. If the end of year is reached then week number is set to 1.



Note: If catchweight is used for this company then, for items with a secondary unit of measure, the secondary stock at the start of the week is updated. The stock at the start of the week is equal to the total physical stock and the stock in-transit, minus all movements in future weeks.

If an item does not have a secondary unit of measure, the secondary stock at the start of the week is zero.



Note: If processing the last week of the year, you should follow this update immediately by month and year end processing since the year number on the Calendar file is only incremented by the year-end update.



When you select the Week End Update option, the Week End Update Confirmation window is displayed containing the Regenerate ABC Classifications field. Enter either 0 (No) or 1 (Yes). Select **Submit=F8** to action or **Exit=F3** to exit without submitting the job.

Month End Update

2/INC

This batch option performs the following tasks for each product/stockroom.

- If the usage cycle set on the company profile is M, for monthly, then Inventory recalculates the average monthly usage.
- Usage this period is set to zero.
- Number of sales orders this period is set to zero.
- Number of substitutions this period is set to zero.
- Current physical stock is stored in a separate field Stock at start of period.
- Inventory calculates the stock value using the appropriate costing method and stored the value in a field called Stock value at start of period.

The option will also:

- Increment the current period number by 1. Set the period number to 1 when you reach the end of the year.
- Produce a stock movements valuation report if this has been requested as part of the submission process. This report is produced based on parameters entered at that time.
- Produces a reorder report as part of the submission process.



Warning: Run Inventory Month End before you run any updates within the Accounts module. Inventory's Calendar dictates the posting of invoices to Accounts Receivable. You must Close Inventory Period to post to a future period, otherwise, postings will continue for the previous period and be incorrect.



Note: If catchweight is used for this company then, for items with a secondary unit of measure, the secondary stock at the start of the period is updated. The stock at the start of the period is equal to the total physical stock and the stock in-transit minus all movements in future periods.

If an item does not have a secondary unit of measure, the secondary stock at the start of the period is zero.

Month End Update Record Report Selection Criteria Screen



Select Month End Update.



Note: When you enter the option, the Stock Movements Valuation Report Parameters screen is displayed.

IN056		NR - Neil Radford and Co. Ltd		User GBJBA1MS		13/08/96	
						16:39:38	
Record Report Selection Criteria							
		From		To			
Stockroom code.....?		█		—			
Item.....?		—		—			
OR							
Item group major/minor.....?		—	—	—	—		
OR							
Item type.....?		—		—			
OR							
Item class.....?		—		—			
Include kits.....	1	1=Include kit components 2=Include kit headers 3=Include kit headers and components					
F3=Exit F4=Prompt							
10-36		SA	MW	KS	IM	II	KENB20A PET166S2

Fields:

Report Parameters

Entry is dependent on what you want to achieve from this routine.

- To action the month end update and print a report, enter a range of parameters and press **Enter** to continue to the Confirmation screen.
- To action the month end update without printing a report, leave all fields blank and press **Enter** to continue to the Confirmation screen.



Press **Enter** to display the Month End Update Confirmation Screen.

Calculate ROP, Max Stock, And EOQ



4/INC

You can calculate the ROP, Maximum Stock and EOQ. These calculations are performed during the Ad-Hoc Update for all items that have 1 entered in the Calculate EOQ, etc. field on the second Stockroom Details Maintenance screen. Select the Ad-Hoc Update option.



*Note: Run the **Ad-hoc Update** after the week/month end update to pick up the latest average usage figures.*

You can see information about the basis of these calculations on the fifth Maintain Company Profile screen, and the second Stockroom Details Maintenance screen.



Select **Confirm Submit=F8** to process the batch job.